

Vote-in-Context: VLMs as Explainable Zero-Shot Rank Fusers

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INTRODUCTION

Information retrieval typically operates in two stages:

- **Retrieval:** Light model(s) propose candidate lists.
- **Reranking:** Heavy model refines the retrieved top-k.

This pipeline powers web search, RecSys, and content retrieval.

PROBLEM STATEMENT

► **The Universality Gap:** Existing rerankers are domain-specific.

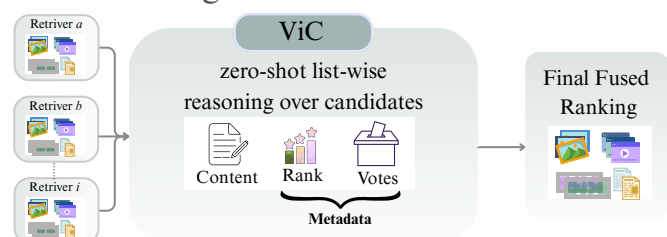
Can we design a single, training-free, and interpretable reranker that handles any modality?

► **The Fusion Gap:** Standard fusion of retrieved lists is content-blind and treated separately from reranking.

Can we extend this framework to perform content-aware fusion, effectively unifying reranking and aggregation?

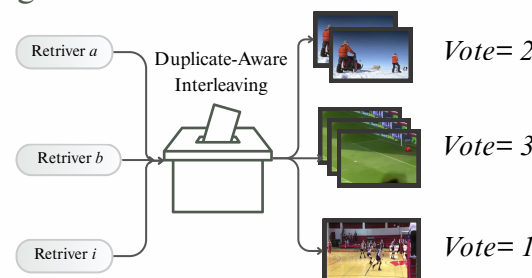
CONTRIBUTIONS

1 **ViC:** A Training-free Universal Reranker & Fuser



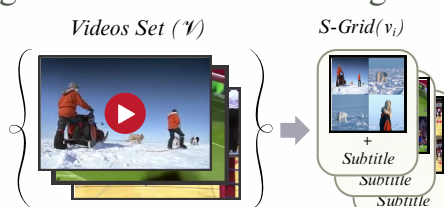
ViC is a framework that transforms ranking into a multimodal reasoning task by processing the top-K retrieved content and its metadata (consensus + rank) in a single zero-shot inference step.

2 **Injecting Votes in Context**



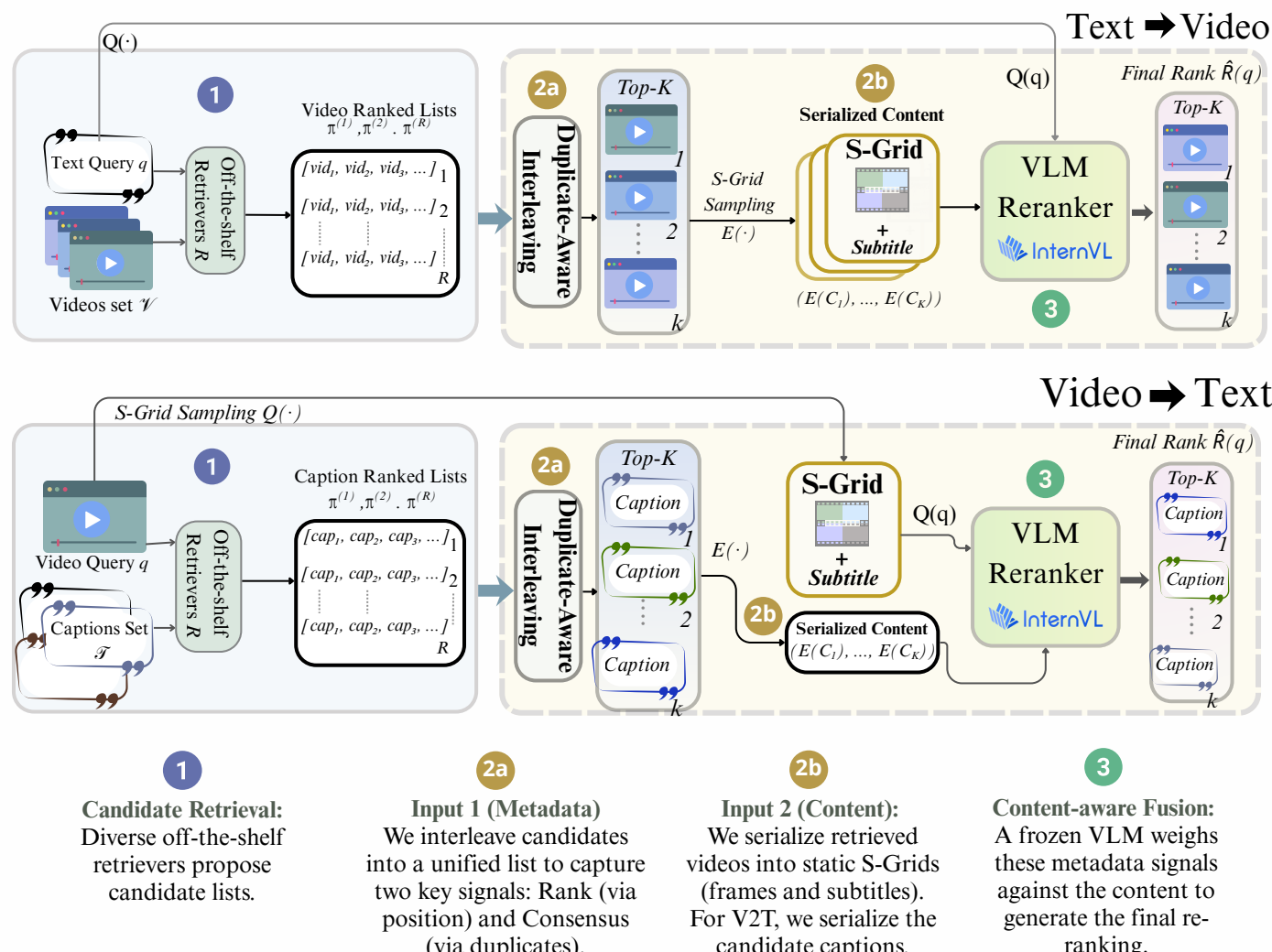
Candidates from different retrievers are interleaved while preserving duplicates to encode cross-retriever consensus within the prompt.

3 **Reducing Video Retrieval to Image Retrieval**

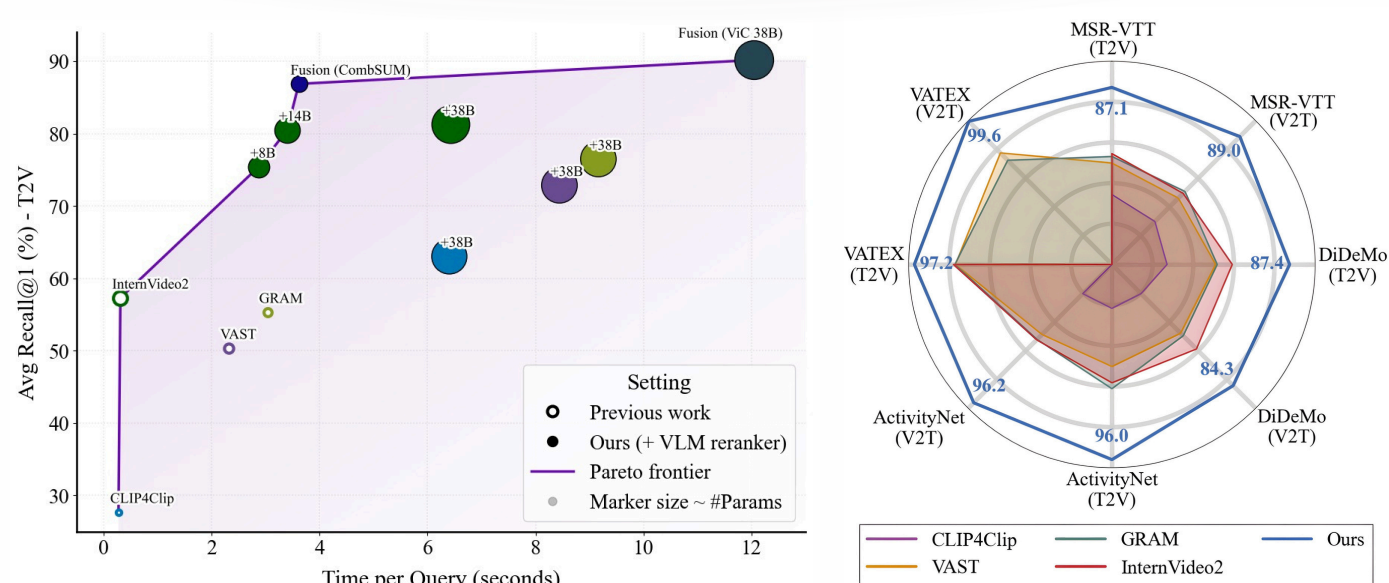


For video applications, we represent content using S-Grid images of tiled frames, with subtitles provided as accompanying text.

APPLYING VIC TO VIDEO RETRIEVAL



RESULTS: EFFICIENCY & GENERALIZATION



ViC redefines the Pareto frontier, delivering massive accuracy gains over previous zero-shot baselines.

ViC (Blue) dominates across all benchmarks, consistently outperforming SOTA in both Text-to-Video and Video-to-Text.

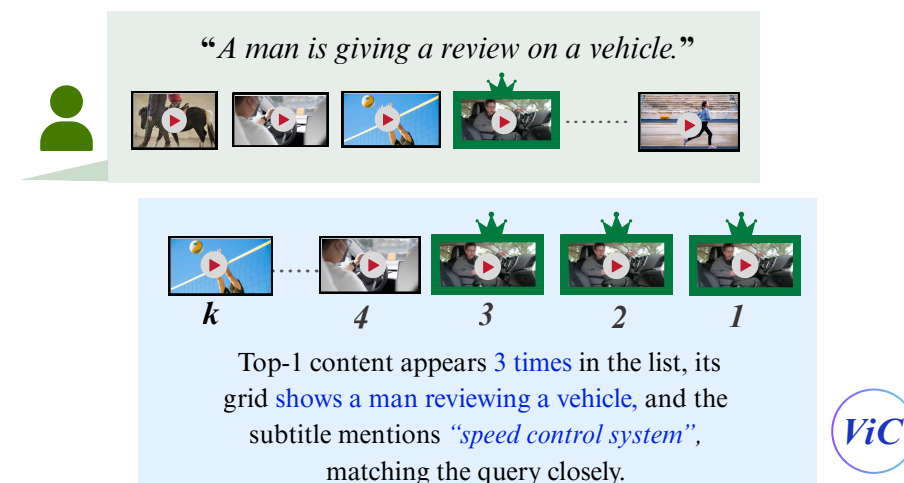
QUANTITATIVE RESULTS

Method \ Benchmark	MSR-VTT	DiDeMo	ActivityNet	VATEX	Average
IV2 (Prev. SOTA)	54.5	59.2	58.2	80.7	63.1
IV2 + ViC	75.9	78.1	89.8	95.8	84.9
RRF (Baseline)	78.3	72.8	96.8	94.7	85.6
ViC (No Duplicates)	84.2	85.5	94.8	96.1	90.1
ViC (Keep Duplicates)	87.1	87.4	96.0	97.5	92.0

● Single Retrieval (ViC as a Reranker) ● Multiple Retrievers (ViC as a Fuser)

Metric used is Recall@1

EXPLAINABLE RERANKING



FUTURE WORK

- **Efficient Reasoning:** A new Pareto frontier would probably emerge from fine-tuning smaller VLMs (<8B).
- **Adaptive S-Grids:** All these benchmarks are for short video retrieval. Long videos probably need a better keyframe selection algorithm rather than random frames.
- **Context Scaling:** Optimizing VLMs to handle larger candidate lists ($K > 30$) without accuracy loss (Currently, $K=14$).

RESOURCES

Paper

Code

Follow Up Paper

